

ARYAN PHILIP

San Diego, CA

+1 (858) 642-5914 arphilip@ucsd.edu LinkedIn GitHub Portfolio

Education

University of California San Diego

Master of Science in Data Science | GPA: 3.9/4.0

September 2023 – June 2025

San Diego, California

Manipal Institute of Technology

Bachelor of Technology in Mechatronics (Robotics) | GPA: 3.87/4.0

July 2018 – April 2023

Bengaluru, India

Experience

University of California San Diego – Causal AI Lab

AI Research Engineer (Advisor: Prof. Biwei Huang)

September 2023 – Present

San Diego, California

- **Shipped Causal Copilot (arXiv:2504.13263)**, a production **LLM-agent system** used by **400+** researchers, by owning the runtime that maps ambiguous analysis requests into plans, tool calls, executable code, statistical checks, and auditable reports.
- Raised held-out workflow completion to **91%** and cut unsupported causal links by **25%** by designing **schema-constrained outputs**, typed tool contracts, deterministic preconditions, verifier passes, retries, and trace-based failure analysis across **20+** methods.
- Reduced structured inference latency by **35%** from 4.2s to under 2.8s by moving agent workflows behind FastAPI services, async batching, cached tool calls, and quantized inference while preserving controllable, reviewable outputs.

Shein Technology

Machine Learning Engineer

June 2024 – August 2024

Palo Alto, California

- Improved high-volume visual policy detection precision by **20%** by fine-tuning **8B/9B vision-language models** and building a CVML decision path for ambiguous product images, compliance signals, and IP-risk categories.
- Built an internal **AI recoloring tool** that converted one product image into up to **10 controlled colorway variants** by combining fine-grained segmentation masks, region selection, color-theory constraints, and texture-preserving recoloring.
- Reduced Photoshop-dependent color iteration by **90%** by giving designers mask refinement, palette comparison, and batch generation controls while balancing latency, precision, and GPU cost under a **4x A100** constraint.

HCL Technologies

Machine Learning Engineer

February 2023 – May 2023

Bengaluru, India

- Raised regulated-domain QA accuracy from **78% to 88%** by building a source-grounded **RAG** service over **100K+** medical documents with retrieval, ranking, citation support, and evidence-constrained generation.
- Reduced retrieval latency by **30%** and end-to-end response time by **50%** by tuning vector indexes, metadata filters, chunk boundaries, re-ranking, and dynamic context selection for production knowledge workflows.
- Made RAG quality measurable by separating retrieval misses, weak ranking, context overload, hallucination risk, and unsupported generation into explicit eval categories that drove targeted fixes instead of generic prompt churn.

Projects

CoMEM-Agent | Python, PyTorch, Multimodal Agent Memory, LoRA, Evaluation Harness

- Improved long-horizon browser-agent reliability by building an evaluation harness where agents read GUI state, choose actions, refresh stale context, recover from tool failures, and emit traceable execution logs.
- Reduced dependence on unbounded prompt histories by designing a **Q-Former + LoRA** memory module that compressed GUI trajectories into **8 continuous memory tokens**.
- Turned agent failures into actionable signals by tracking verifier refreshes, action recovery, constrained inference, stale-state errors, and failure-mode summaries across public web-agent suites.

Traffic Flow Forecasting | Python, C++, PyTorch, GNNs, Plotly Dash

- Improved 60-minute highway forecasting MAE by **18%** by training diffusion-convolutional **GNNs** on live Caltrans sensor data and pairing vectorized **C++** preprocessing with an inspection dashboard.

Technical Skills

Languages: Python, C++, SQL, TypeScript, Bash

Agentic AI / ML: Planning, Tool Calling, Structured Outputs, RAG, Agent Evaluation, Tracing, Failure Handling, LLM-Assisted Coding, PyTorch, VLMs, LoRA, Quantization, GNNs

Systems: FastAPI, Docker, MLflow, Vector Databases, Telemetry, AWS, GCP, Azure, CI/CD, Git